

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2018

EC 309.01/EC 309 Digital Communication

Date: 03.05.2018, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 A zero memory source emits six messages with probabilities 0.3, 0.25, 0.15, 0.12, 0.1, [04]

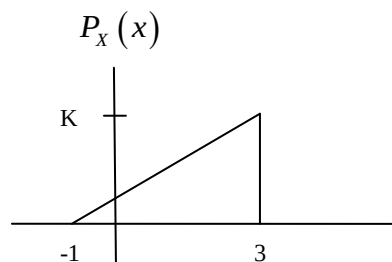
(a) and 0.08 respectively. Find the 4-ary Huffman code. Determine the average word length, the efficiency and the redundancy.

(b) In a random experiment, a trial consists of three successive tosses of a coin. If we [03]
define an RV X as the number of heads appearing in a trial, determine $P_X(x)$
and $F_X(x)$.

(C) State and prove sampling theorem in time domain. [04]

Q-2 Attempt any two [12]

(a) Find the mean, the mean square, and the variance of the RV x whose PDF is shown in [06]
figure.



(b) A zero memory source emits messages m_1 and m_2 with probabilities 0.8 and 0.2 [06]
respectively. Find the optimum binary code for this source as well as for its second
order extensions. Determine the code efficiencies in each case.

(c) Shows that the output SNR increases exponentially with the bandwidth expansion [06]
factor B_T/B .

Q-3 Attempt any two [12]

(a) A Binary Channel matrix is given by [06]

$$\begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{10} & \frac{9}{10} \end{bmatrix}$$

This means

$$P_{y/x}(y_1/x_1) = 2/3, P_{y/x}(y_2/x_1) = 1/3, \text{ etc.}$$

$$P_x(x_1) = 1/3 \text{ and } P_x(x_2) = 2/3. \text{ Determine } H(X), H(X/Y), H(Y), H(Y/X) \text{ and } I(X; Y)$$

(b) Explain the ISI and signal degradation study using eye diagram. [06]

(c) Explain the need of non uniform quantization and describe the companding in detail. [06]

SECTION - II

Q-4 [03]

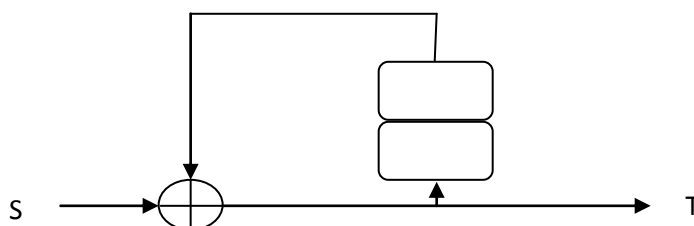
Compare PAM, PPM and PWM.

(a) [02]

(b) Data at a rate of 6 kbit/s is to be transmitted over a leased line of bandwidth 4 kHz using Nyquist criterion pulses. Determine the maximum value of roll –off factor r that can be used. [02]

(c) A binary source generates digit 1 and 0 randomly with equal probability. Assign probabilities to the following events with respect to 10 digits generated by the source: (a) there are exactly two 1's and eight 0's, (b) there are at least four 0's. [02]

(d) A scrambler is shown in figure. If the sequence $S = 101010100000111$ is applied to the input of this scrambler determine the output sequence T . [04]



Q-5 Attempt any two [12]

(a) What are the desirable properties of line coding? Also give comparative study of different line codes. [06]

(b) Explain Quantization. [06]

(c) In a Binary data transmission using duobinary pulses, sample values of the received pulses were read as follows. (First Bit in the sequence is one) [06]

1 2 0 -2 -2 0 0 -2 0 2 0 0 2 0 0 0 -2

1. Explain if there is any error in detection

2. If there is no detection error, determine the received bit sequence.

Q-6 Attempt any two [12]

(a) In certain telemetry system, there are eight analog instruments, each of bandwidth 2 kHz. Sample of these signals are time-division multiplexed, quantized, and binary coded. The error in sample amplitudes cannot be greater than 1% of the peak amplitude. [06]

1. Determine L, the number of quantization levels.

2. Find the transmission bandwidth B_T if nyquist criterion pulses with roll off factor $r = 0.2$ are used. The sampling rate must be at least 25% above the nyquist rate.

(b) Explain the generation of coherent binary frequency shift keying (BPSK) modulation technique. Draw the constellation diagram for same. [06]

(c) Explain the generation of coherent binary frequency shift keying (BFSK) modulation technique. Draw the constellation diagram for same. [06]
